# 3 REVOLUTIONISING ROBOTICS: STARTUP UNLEASHES THE POWER OF ROS USING HARDWARE ACCELERATION

In an era where robotics is rapidly transforming industries and reshaping the way we interact with technology, one company stands at the forefront of this revolution by harnessing the power of the robot operating system (ROS) using hardware acceleration

Acceleration Robotics is a startup focused on designing customised hardware, or the 'brains' that speed up a robot's operation. In particular, the company creates custom compute architectures for high-performance robots through hardware acceleration solutions (CPUs, FPGAs, GPUs, and combinations). By leveraging the flexibility, modularity, and collaborative nature of ROS, they are revolutionising the way robotic systems are designed, developed, and deployed.

"Robot operating system (ROS) is a set of software libraries and tools that you can use to build your robot applications," says Prateek Nagras, CEO of Acceleration Robotics.

Acceleration Robotics aims to simplify and democratise hardware acceleration in robotics development. While discussing the challenges in the field of robotics, Prateek says that "there is a big gap between the semiconductor industry, which is basically manufacturing the FPGAs and GPUs and CPUs, and the robotics industry, where they consume these chips to unveil their own software stacks on top of it."

At the core of Acceleration Robotics' approach lies the power of ROS, an open source framework that provides a standardised and flexible platform for building and controlling robots. By embracing ROS as its foundation, Acceleration Robotics benefits from the modular architecture, hardware abstraction, and interoperability that ROS offers. This allows them to focus on developing high-level robotic functionalities while seamlessly integrating with various hardware components and third-party tools. By harnessing the modularity and reusability of ROS, they can rapidly prototype and iterate their robotic systems, significantly reducing development time and costs.

Acceleration Robotics is working with Peer Robotics in bringing hardware acceleration to their mobile robotics software stack

Acceleration Robotics is one of the first companies in the country to have started working on ROS 2. This agility enables them to stay at the forefront of technological advancements, ensuring that the robots of their clients are equipped with the latest capabilities and features. The hardware independence provided by the ROS allows Acceleration Robotics to adapt its solutions to different robot platforms, expanding its market reach and catering to diverse customer needs.

Prateek Nagras, the CEO of Acceleration Robotics

While discussing a project, Prateek talks about the Robot Core Framework developed by their company, which aims to build custom compute architectures for robots using a vendor-agnostic approach. He explains that they have worked with the ROS 2 framework and have contributed to the ROS enhancement proposal (REP) process to standardise the framework. This standardisation allows roboticists to easily use different accelerators (GPUs, FPGAs, etc) by changing just a few lines of code.

Acceleration Robotics has already built a framework and is working on new IP courses, which are like a proprietary piece of software for advanced use cases such as accelerating software stacks like perception, navigation, and manipulation. The company has worked with multiple companies such as AMD, Intel, and Peer Robotics, to name a few.

On discussing their future plans, Prateek says, "Right now we are just concentrating on software and use off-the shelf hardware, but maybe in a few years, we could even start building our own custom hardware dedicated for robots, so we can basically have our own hardware plus software combination." EFY

—*Sharad Bhowmick*